

ROBIN INHO KEE

Department of Robotics, University of Michigan, Ann Arbor, MI

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Education

University of Michigan

Doctor of Philosophy in Robotics

- Recipient of the Draper Scholar Program, a full Ph.D. fellowship in autonomy (Advisor: Prof. Vasileios Tzoumas)

Aug 2025 - Present

Ann Arbor, MI, USA

University of Michigan

Master of Science in Mechanical Engineering

Aug 2023 – Dec 2024

Ann Arbor, MI, USA

Yonsei University

Bachelor of Science in Mechanical Engineering | Military Service: Oct 2020 – Jul 2022

Mar 2016 – Feb 2022

Seoul, South Korea

Research Interest

Multi-Robot Autonomy, Safety-Critical Learning-Based Control, Uncertainty-Aware Planning, Scalable Decision-Making

Publications

***: Equally Contributed**

Under review

1. Sangdo Kim*, Jeonguk Kang*, **Robin Inho Kee**, Sunwoo Kim, Choa Kim, Youngsu Cha, Yisoo Lee, Kanggeon Kim, Jongwon Lee, “Towards Daily Life Sarcopenia Detection: Deep Learning-Based Gait Analysis Using Wearable Hip Assistive Robot”, *Under Review* [Project Page]
2. Jin-Hee Lee, Dahyun Nam, **Robin Inho Kee**, YoungKey Kim, Seok-Jun Buu, “Vision-Ultrasound Robotic System based on Deep Learning for Gas and Arc Hazard Detection in Manufacturing”, *Under Review* [arxiv]

Journals

1. **Robin Inho Kee**, Dahyun Nam, Seok-Jun Bu, Sung-Bae Cho, “Disentangled Prototyping with Triplet-trained Prototypical Network for Few-shot Learning in In-vehicle Noise Classification”, *IEEE Access*, 2024 [Paper] [Project Page]
2. Hobin Kim, Jongbok Lee, Sunwoo Kim, **Inho Kee**, Sangdo Kim, Shinsuk Park, Kanggeon Kim, Jongwon Lee, “Gait Phase Estimation Method Adaptable to Changes in Gait Speed on Level Ground and Stairs”, *The Journal of Korea Robotics Society*, 2023
3. Byonghun Kim, Sunghyun Hong, Inwook Oh, Yangwoo Lee, **Inho Kee**, Saeyong Lee, “Measurement of ankle joint movements using IMUs during running”, *Sensors*, 2021 [Paper] [Project Page]

Conferences

1. **Robin Inho Kee***, Taehyeun Kim*, Anouck Girard, Ilya Kolmanovsky, “Time Shift Governor-Guided MPC with Collision Cone CBFs for Safe Adaptive Cruise Control in Dynamic Environments”, *IEEE Conference on Control Technology and Applications (CCTA)*, 2025 [Project Page]
2. Taekyung Kim, **Robin Inho Kee**, Dimitra Panagou, “Learning to Refine Input Constrained Control Barrier Functions via Uncertainty-Aware Online Parameter Adaptation”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025 [arxiv] [Project Page] [Github]
3. Taehyeun Kim*, **Robin Inho Kee***, Ilya Kolmanovsky, Anouck Girard, “Constrained Control for Autonomous Spacecraft Rendezvous: Learning-based Time Shift Governor”, *AIAA SciTech Forum*, 2025 [Paper] [arxiv] [Project Page]
4. Dahyun Nam, **Inho Kee**, Seok-Jun Bu, SungBae Cho, “Dynamic Prototype-guided Memory Replay for In-Vehicle Noise Classification”, *Korea Data Mining Society*, 2023, **SAS Student Paper Award** [Project Page]
5. Woojin Jo, Sehyun Hwang, **Inho Kee**, Soohong Lee, “An Intelligent Lock-Out Tag-Out System for Monitoring and Control of the Locked Device”, *IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 2019 [Project Page]

Research Experiences

Graduate Student Research Assistant, University of Michigan

Aug 2025 – Present

Intelligent Robotics and Autonomy Lab (PI: Prof. Vasileios Tzoumas)

- Working on resilient multi-robot autonomy, focusing on decentralized, distributed, communication-aware safe exploration and uncertainty-aware control frameworks

Graduate Research Assistant, University of Michigan

May 2024 – Aug 2025

Distributed Autonomous Systems and Control Lab (PI: Prof. Dimitra Panagou)

Ann Arbor, MI, USA

- Developed and implemented the Probabilistic Ensemble Neural Network (PENN) for real-time adaptation of Input Constrained Control Barrier Functions (ICCBF) [Project Page]
- Designed and integrated a two-step uncertainty verification process using Jensen-Rényi Divergence (JRD) and distributionally robust Conditional Value at Risk (CVaR) to ensure model confidence and local validity

Graduate Research Assistant, University of Michigan

Jan 2024 – Feb 2025

Vehicle Optimization, Dynamics, Control and Autonomy Lab (PI: Prof. A. Girard and I. Kolmanovsky) Ann Arbor, MI, USA

- Developed a learning-based Time Shift Governor (L-TSG) to address computational challenges in spacecraft rendezvous and docking under dynamic constraints in elliptic orbits and the SpaceX Crew3 mission, integrating a constraint-guided recurrent neural network, achieving 10x faster computation while maintaining robustness [\[Project Page\]](#)
- Designed and implemented a TSG-guided MPC-CBF framework for adaptive cruise control (ACC) on curved roads, enabling real-time adaptation to dynamic uncertainties like sudden braking or lead vehicle reversals, enhancing robustness [\[Project Page\]](#)

Researcher, Yonsei University

Mar 2023 – Aug 2023

Soft Computing Lab (PI: Dr. Seok-Jun Bu)

Seoul, South Korea

- Engineered a disentangled prototypical convolutional network for advanced in-vehicle noise classification, enhancing few-shot learning in automotive acoustic analysis with accuracy of 96.81% on a 9-way 1-shot task [\[Project Page\]](#)
- Presented novel in-vehicle noise classification deep learning model using dynamic prototype-guided memory replay method demonstrating 6.38% improvement in accuracy

Research Intern, Korea Institute of Science and Technology

Jul 2022 – Jul 2023

Assistive and Interactive Robotics Lab (PI: Dr. Jongwon Lee)

Seoul, South Korea

- Improved wearable hip complex assistive robot with 4DOF active joint [\[Project Page\]](#)
- Developed a deep learning model for estimating foot trajectory by fusing data from hip exoskeleton and insole sensors, achieving 100% accuracy in identifying sarcopenia patients through gait parameter analysis.
- Led and administered motion capture system (Motion Analysis) experiments over 40 subjects, including patients and outdoor hiking experiments over 200km

Research Engineer, Yonsei University

May 2020 – May 2021

Integrative Sports Science Research Lab (PI: Prof. Byong Hun Kim)

Seoul, South Korea

- Initiated and developed a portable real-time ankle angle analysis audio-visual feedback system [\[Project Page\]](#)
- Validated IMU measurement of joint kinematics against Vicon system using a developed wearable device
- Conceptualized subtalar joint angle estimation algorithm with random forest method

Research Intern, Seoul National University

Apr 2020 – Oct 2020

Innovative Design and Integrated Manufacturing Lab (Advisor: Prof. Sung-Hoon Ahn)

Seoul, South Korea

- Developed lab automation (tensile test) using an autonomous mobile manipulator
- Customized modular mechanical end effector implemented with torque and position controller
- Devised low-cost appropriate robotic manipulator (Open quasi-direct drive robot)

Selected Awards and Honors

Draper Scholars Program (Full Ph.D. Fellowship), Charles Stark Draper Laboratory, 2025–Present
SAS Student Paper Award, Conference of Korea Data Mining Society, Korea Data Mining Society, 2023
1st Place, International S.M.A.R.T Startup Competition, Innovative Technology and Energy Center, 2020
Science and Engineering Undergraduate Internship Program Scholarship, Yonsei University, 2020
Academic Highest Honors, Yonsei University, 2019, 2020
2nd Prize, Robot Open Source Lab, Samsung Open Source Conference, 2019
Academic Excellence Scholarship, Yonsei University, 2018, 2019

Extracurricular experience

Roboin, Robotics club

Jun 2018 – Aug 2023

Advisory Committee (2020-2023), President (2019)

Yonsei University, Seoul, South Korea

- Led club-wide technical seminars, interdisciplinary team projects, and national competitions on robotics
- Directed autonomous robotics projects:
 - * Autonomous fire-extinguishing robot with heat-source detection and navigation [\[Project Page\]](#)
 - * Teo-Jansen legged robot with OpenCV and tiny-YOLOv4-based navigation [\[Project Page\]](#)

Military Service

1st Fighter Wing, Republic of Korea Air Force

Oct 2020 – Jul 2022

Staff Sergeant, Military Intelligence Airman

Gwangju, South Korea

- Received honorable discharge and recognized as the top sharpshooter among over a thousand peers
- Demonstrated leadership and decision-making under pressure, contributing to national security analysis